WASP-10b

R-0851

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-10_230906` · created 2026-07-28T20:49:01+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460193.8325 +/- 0.006 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.036 +/- 0.06
Transit depth (Rp/Rs)^2	0.13 +/- 0.44 [%]
Orbital Inclination (inc)	83.61 +/- 1.94
Airmass coefficient 1 (a1)	31.63 +/- 24.16
Airmass coefficient 2 (a2)	-3.0 +/- 1.01
Scatter in the residuals of the lightcurve fit is	113.21 %
Best Comparison Star	None
Optimal Aperture	2.29
Optimal Annulus	3.0
Transit Duration (day)	0.014 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.036 ± 0.06 vs 0.1592 (deviation 0.98σ)
Duration fit vs published	0.014 d vs 0.0946 d (deviation 1.54σ)
Mid-time O–C	-4.6 min
Depth SNR	0.3
Residual scatter	113.21 %

Light curve

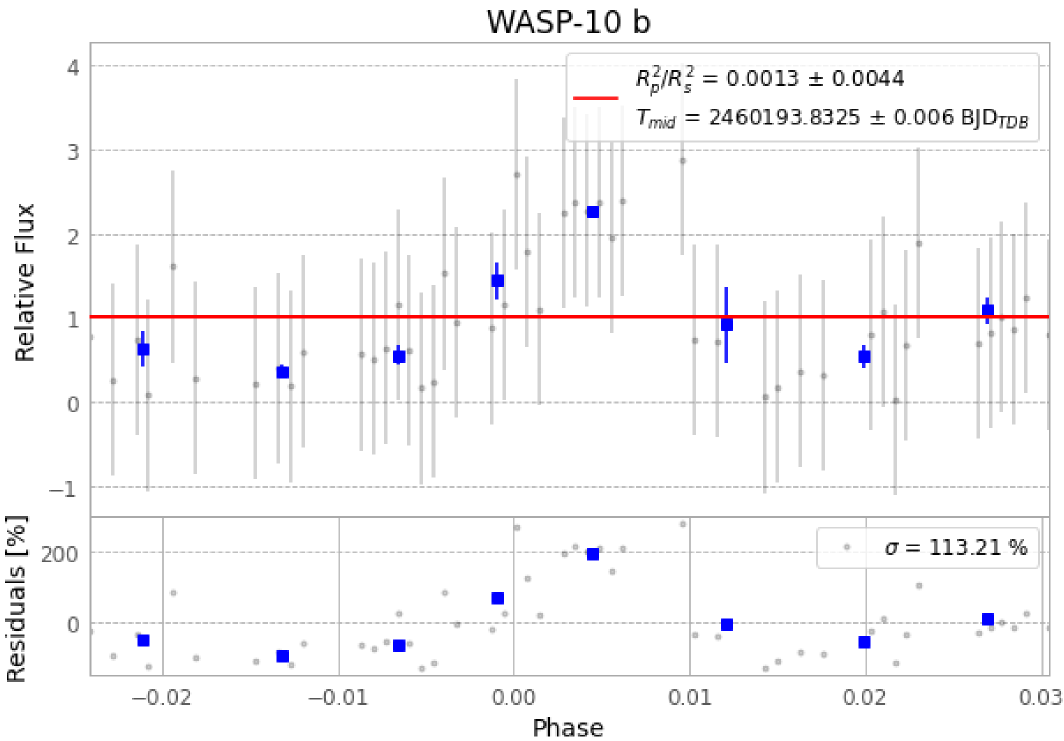

Figure 1 — FinalLightCurve_WASP-10 b_2023-09-05.png (SHA-256 8835f7bffa372f27…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [109, 265], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 39363f0820b34461…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

83 FITS frames (55 MB), `WASP-10230906060015.FITS` → `WASP-10230906100615.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-230906.log (b14213631cc2…), FinalLightCurve_WASP-10 b_2023-09-05.png (8835f7bffa37…), FinalLightCurve_WASP-10 b_2023-09-05.csv (8319393c22a2…)

Compute resources

Wall time 135.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)